Integrating Along Curves and Across Surfaces: Densities and Forms

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A wire carrying mass and a force acting along the same wire lead to different integrals. Mass accumulates according to length, while work is sensitive to the direction of motion. The analogous distinction on a surface is between accumulated scalar density and signed flux. The formulas become much easier to remember once we derive them from parametrizations and recognize that only the oriented integrals are differential forms.[1]

§1.1 Length versus Displacement Integrals

Let $\gamma : [a, b] \to \mathbb{R}^3$ be a continuously differentiable regular curve. A small parameter interval of length dt corresponds to arc length

$$ds = \|\gamma'(t)\|\, dt.$$

Thus a scalar density ρ distributed along a wire has total mass

$$\boxed{\int_\gamma \rho\, ds = \int_a^b \rho(\gamma(t))\|\gamma'(t)\|\, dt.}$$

The norm is essential: it makes the contribution nonnegative when the orientation is reversed.

A force field F performs signed work

$$\boxed{\int_\gamma F \cdot d\mathbf{r} = \int_a^b F(\gamma(t)) \cdot \gamma'(t)\, dt.}$$

Its integrand is a signed projection onto the displacement direction. Traversing the path backward changes the sign.

On a circle $\gamma(t) = (R\cos t, R\sin t)$, $0 \le t \le 2\pi$, a linear density $\rho = x^2 + y^2$ gives

$$\int_\gamma \rho\, ds = 2\pi R^3.$$

For the tangential field $F = (-y, x)$, however,

$$F(\gamma(t)) \cdot \gamma'(t) = R^2, \qquad \int_\gamma F \cdot d\mathbf{r} = 2\pi R^2.$$

The distinction can be observed physically: mass does not change when the wire is traversed in the opposite direction; work does.

[1] MIT 18.02, *Surface Integrals*, supplementary note V9.

§1.2 Why Arc-Length Uses an Absolute Value

Suppose $\widetilde{\gamma}(s) = \gamma(\phi(s))$, where ϕ is a smooth monotone bijection of parameter intervals. Then

$$\widetilde{\gamma}'(s) = \gamma'(\phi(s))\phi'(s).$$

For length, the speed transforms with $|\phi'(s)|$. The absolute value compensates for any reversal of parameter direction, leaving the mass integral unchanged.

For work, the derivative $\phi'(s)$ retains its sign. If ϕ reverses the interval, the work integral changes sign. This difference between absolute and signed Jacobian factors is the local reason the two line-integral formulas have different orientation behavior.

The work integral can be written using the differential one-form

$$\alpha = F_1\,dx + F_2\,dy + F_3\,dz.$$

Along γ, its pullback is

$$\gamma^*\alpha = \big(F(\gamma(t)) \cdot \gamma'(t)\big)\,dt.$$

Thus $\int_\gamma \alpha = \int_{[a,b]} \gamma^*\alpha$. Pullback explains the whole calculation without requiring the parameter to have any intrinsic physical meaning.

§1.3 Area is the Gram determinant of the tangent vectors

For a smooth parametrized surface $r : D \subset \mathbb{R}^2 \to \mathbb{R}^3$, the vectors r_u, r_v span a tangent parallelogram. Its area is

$$\|r_u \times r_v\|\,du\,dv = \sqrt{\langle r_u, r_u\rangle\langle r_v, r_v\rangle - \langle r_u, r_v\rangle^2}\,du\,dv.$$

The square root is the Gram determinant of the induced metric on the parameter domain. Therefore the scalar surface integral is

$$\boxed{\iint_S f\,dS = \iint_D f(r(u,v))\|r_u \times r_v\|\,du\,dv.}$$

Like arc length, area is insensitive to a reversal of orientation.

For the lateral surface of a cylinder of radius R and height H, take

$$r(\theta, z) = (R\cos\theta, R\sin\theta, z), \quad 0 \le \theta \le 2\pi, \quad 0 \le z \le H.$$

Since $\|r_\theta \times r_z\| = R$, the integral of $f(x,y,z) = z^2$ is

$$\iint_S z^2\,dS = R\int_0^{2\pi}\int_0^H z^2\,dz\,d\theta = \frac{2\pi R H^3}{3}.$$

The calculation is just length around the circumference times the integral of density over height, but the Gram factor makes it independent of which parameters describe the cylinder.

§1.4 Flux uses the oriented area vector

A flux integral measures the component of a vector field F crossing the chosen side of a surface:

$$\iint_S F \cdot n \, dS = \iint_D F(r(u,v)) \cdot (r_u \times r_v) \, du \, dv,$$

provided $r_u \times r_v$ gives the prescribed orientation.

For the same cylinder, $r_\theta \times r_z = (R\cos\theta, R\sin\theta, 0)$ points outward. Let $F = (x, y, 0)$, so the field points radially away from the axis. Then

$$F(r) \cdot (r_\theta \times r_z) = R^2, \qquad \boxed{\iint_{\text{lateral } S} F \cdot n \, dS = 2\pi R^2 H.}$$

The top and bottom caps contribute zero because F is horizontal. Since $\operatorname{div} F = 2$ and the cylinder's volume is $\pi R^2 H$, the divergence theorem gives the same total flux $2\pi R^2 H$. The direct parametrization and the volume calculation provide independent checks of the orientation and scale.

§1.5 A vector field gives a one-form and a two-form

In oriented Euclidean $\mathbb{R}^3$, the vector field $F = (P, Q, R)$ gives a one-form

$$\alpha_F = P \, dx + Q \, dy + R \, dz$$

and a two-form

$$\beta_F = P \, dy \wedge dz + Q \, dz \wedge dx + R \, dx \wedge dy.$$

The one-form measures work along tangent vectors; the two-form measures flux across oriented tangent parallelograms.

These are related by the Euclidean Hodge star:

$$\boxed{\beta_F = \star \alpha_F.}$$

The star depends on the metric and orientation. It encodes the operation of converting a tangent direction into the complementary oriented area direction.

For a parametrization $r(u, v)$, the two-form pulls back to

$$r^* \beta_F = \big(F(r(u,v)) \cdot (r_u \times r_v)\big) \, du \wedge dv.$$

The vector-calculus flux formula is exactly this pullback formula.

Exterior differentiation then reproduces the two standard differential operators:

$$d\alpha_F = (\operatorname{curl} F) \cdot d\mathbf{S}, \qquad d\beta_F = (\operatorname{div} F) \, dx \wedge dy \wedge dz.$$

Here $(\operatorname{curl} F) \cdot d\mathbf{S}$ means the two-form built from the curl by the same rule used for β_F. Integrating these identities yields the familiar Stokes and divergence theorems as different degree cases of $\int_{\partial M} \omega = \int_M d\omega$.

§1.6 Nonorientability is a genuine obstruction

The factor $\|r_u \times r_v\|$ makes scalar surface integration possible even on a Möbius strip, because local area elements agree after changing charts. To define signed flux, however, the normal must vary continuously and consistently across the entire surface. A Möbius strip has no global continuous choice of unit normal.

One can still integrate a scalar density over it, and a chosen local patch can carry flux relative to a local normal. But a global signed flux through the whole Möbius strip has no canonical meaning for a general field. The failure is topological: the obstruction remains even with arbitrarily accurate parametrization formulas.

This separates the roles of metric, orientation and topology. The metric supplies lengths and areas. Orientation supplies the signs for work and flux. Topology determines whether an orientation can be chosen globally. Once those roles are clear, the first-kind and second-kind integration formulas follow naturally from the geometry of what is being accumulated.